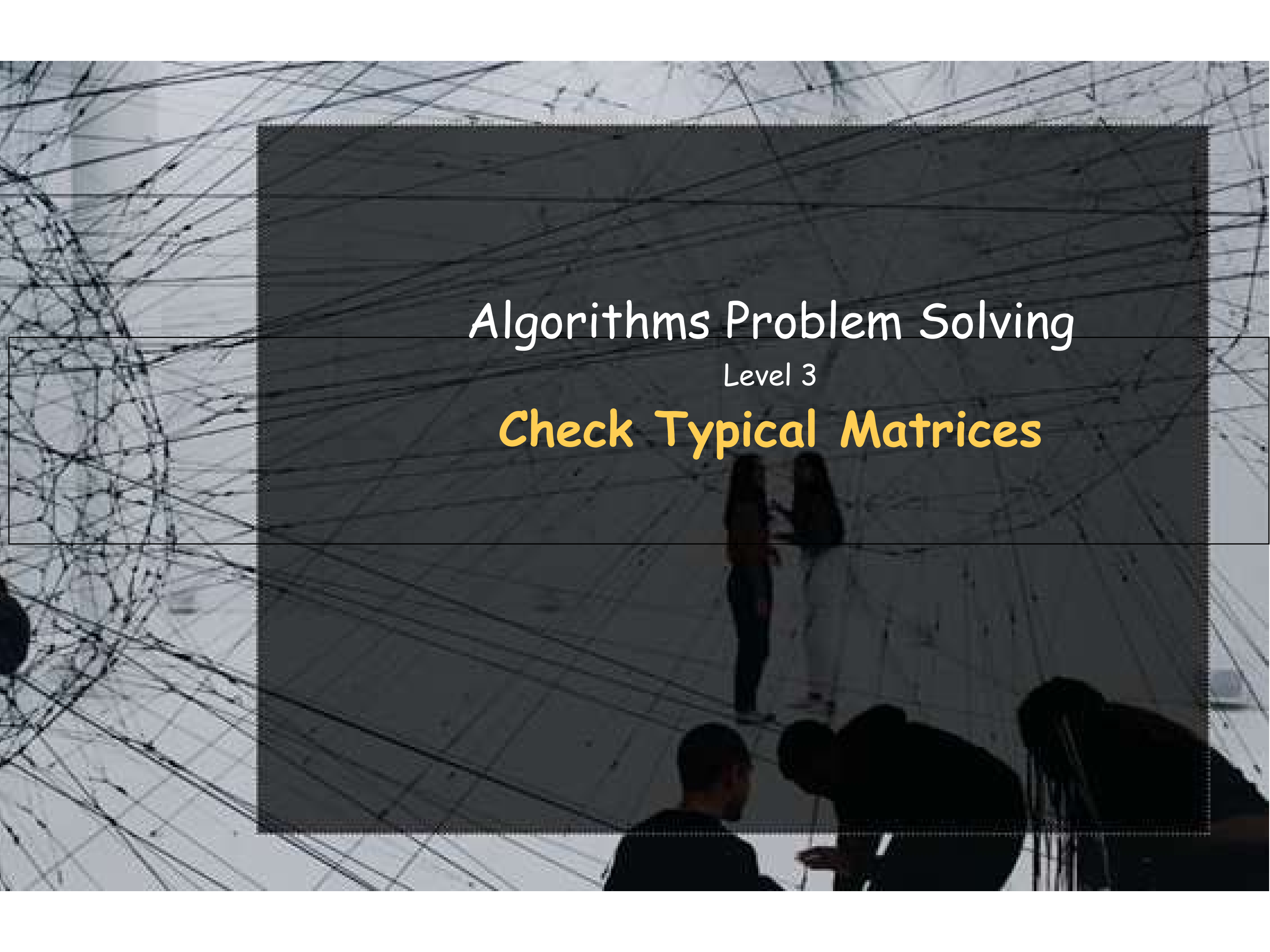


Algorithms Problem Solving

Level 3

Check Typical Matrices



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Definition

Structural and Content Matrix Equality

 Program checks whether two matrices contain exactly the same values at exactly the same positions.

 True matrix equality requires matching dimensions and matching corresponding elements.

Explanation

Element-by-Element Comparison

-  The program traverses both matrices using nested loops.

-  Each `matrix1[i][j]` is compared directly with `matrix2[i][j]`.

Early Exit

-  As soon as one pair of corresponding elements differs, the function returns `false`.

-  There is no need to inspect the remaining elements after a mismatch has already been found.

Exact Equality

-  Unlike sum-based comparison, this method verifies the actual structure and contents of the matrices.

-  Two matrices are considered identical only when every corresponding element matches.

Procedures

1. Create or initialize two matrices.
2. Display both matrices.
3. Call ``AreMatricesIdentical()``.
4. Compare corresponding elements using nested loops.
5. Return ``false`` immediately when a mismatch is found.
6. Return ``true`` if all elements match.

💡 Example

Matrix1:

6 3 9

4 5 9

3 4 4

Matrix2:

8 9 6

6 8 6

9 3 5

➡ Result

No: matrices are NOT equal.

Note

- ◆ Early exit avoids unnecessary comparisons after the first mismatch.
- ◆ The matrices must have compatible dimensions before position-by-position comparison.
- ◆ For floating-point matrices, exact `==` comparison may be inappropriate because of precision errors.

⊘ Warning

- ◆ Do not confuse identical matrices with matrices that merely have equal sums.
- ◆ Returning `false` only after checking every element is less efficient than stopping at the first mismatch.
- ◆ Floating-point equality may require an epsilon tolerance such as `abs(a - b) < epsilon`.



Comparison

▣ `AreMatricesIdentical()`

- ✓ Compares corresponding elements directly.
- ✓ Stops immediately when a mismatch is found.
- ✓ Provides exact content comparison when the dimensions match.

▤ `AreMatricesEqual()` by Sum

- ✓ Compares only aggregate totals.
- ✓ Cannot detect different element arrangements with equal sums.
- ✓ Must calculate the complete sum before comparison.

✦✦ Summary

- ✓ Traverses two matrices position by position.
- ✓ Verifies both value and position equality.
- ✓ Uses early return to avoid unnecessary work.
- ✓ Provides a true element-based matrix equality check.

Thank you 😊